



POTSDAM INSTITUTE FOR
CLIMATE IMPACT RESEARCH

GAMS code, modules & realizations

MAGPIE Workshop 2026

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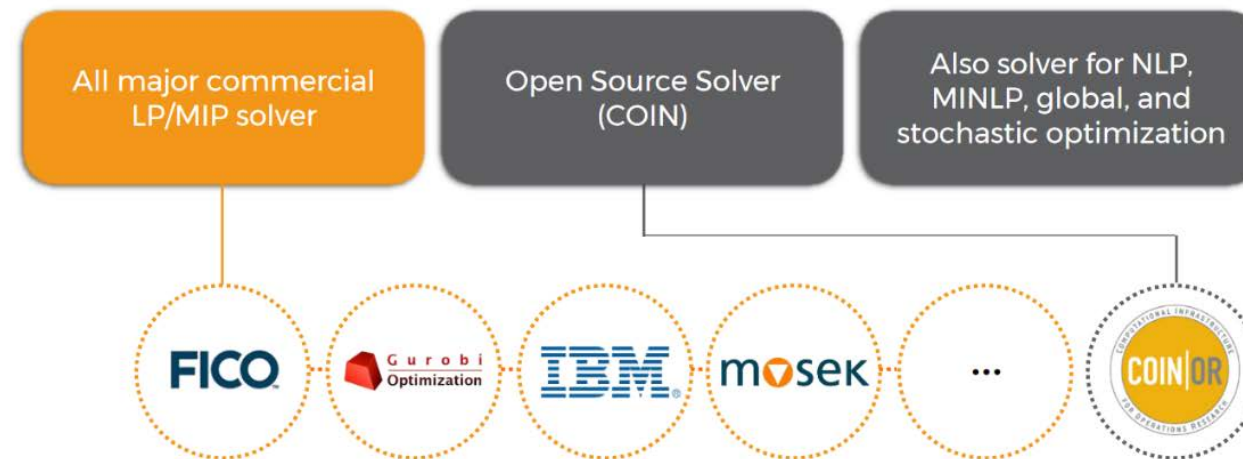
June 16, 2026

- What is GAMS?
- General structure of the MAgPIE Model
- **The magpie folder:** Components of MAgPIE, the structure of modules and realizations, and the full.gms file
- Coding etiquette: Variable and parameter naming
- MAgPIE environment: pre- and postprocessing
- Brief exercise

What is GAMS?

- › MAgPIE is written in the **General Algebraic Modeling System (GAMS)** language

*“GAMS is a high level modeling system for **mathematical programming** and **optimization**. It consists of a **language compiler** and a range of associated **solvers**.”*



Useful links:

GAMS at a glance: <https://www.gams.com/products/gams/gams-language/>

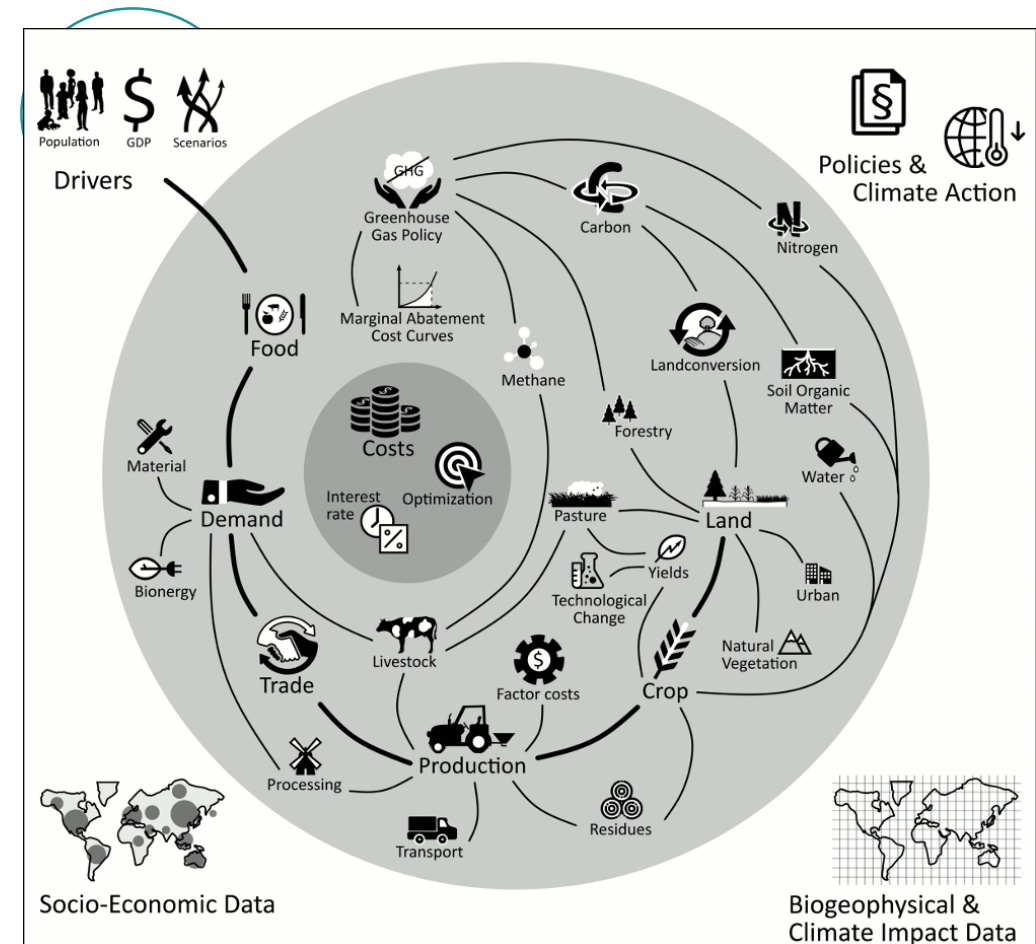
GAMS documentation: <https://www.gams.com/latest/docs/>

General structure of the MAgPIE Model



- Future **population** and **GDP** are the main drivers (different **scenarios**).
- **Food consumption** patterns lead to **demand** for primary agricultural products.
- **Trade** patterns lead to regional **production**.
- **Production and biophysical** yield and water availability data shape **cropping patterns**.
- **Cropping patterns** drive **land use** decisions.

These interactions and calculations are described in the modules of MAgPIE.

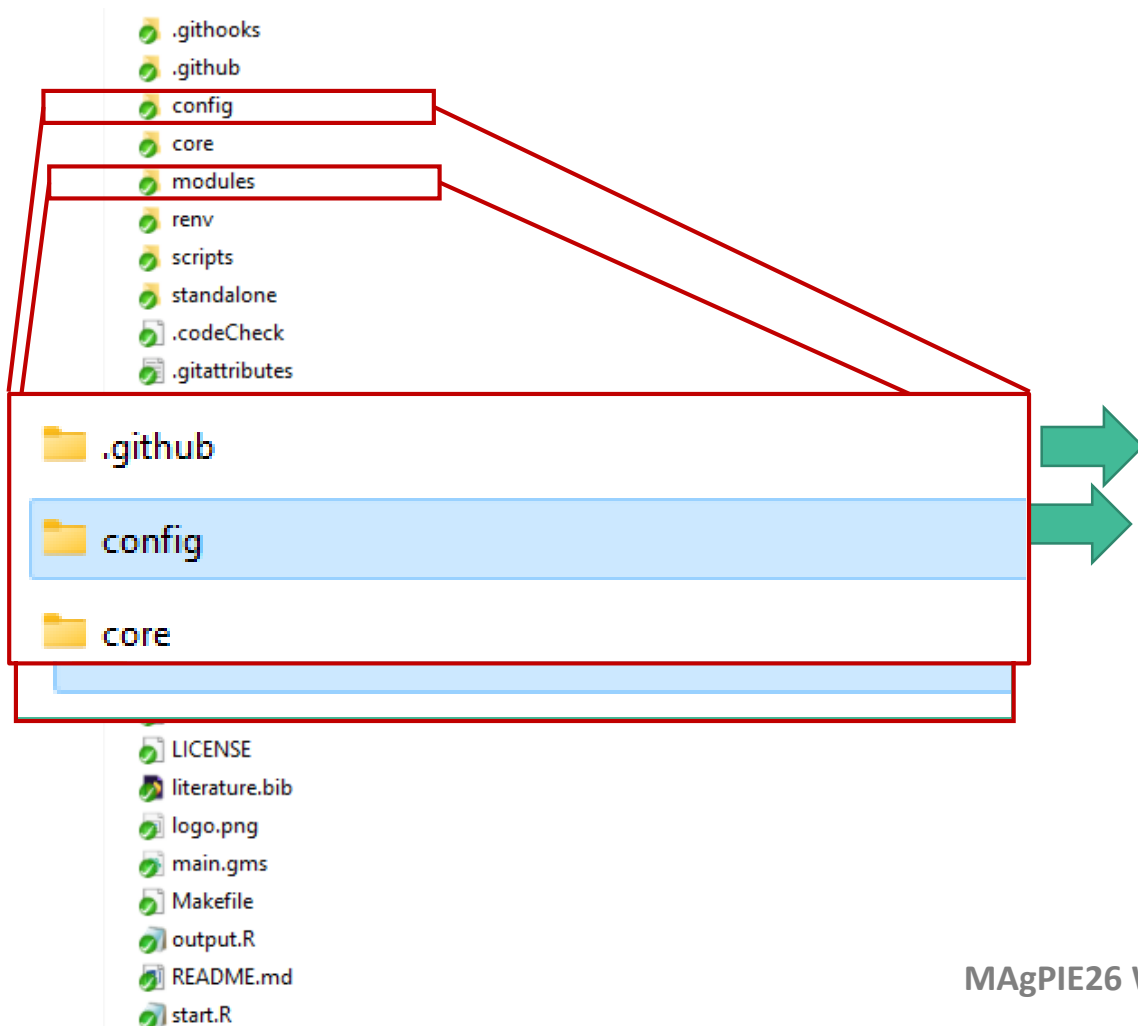


More info at: <https://rse.pik-potsdam.de/doc/magpie/4.14.0/>

The magpie folder: Components of MAgPIE



After you have cloned the magpie folder (*git clone <https://github.com/magpiemodel/magpie.git>*) from the repository, you will find :



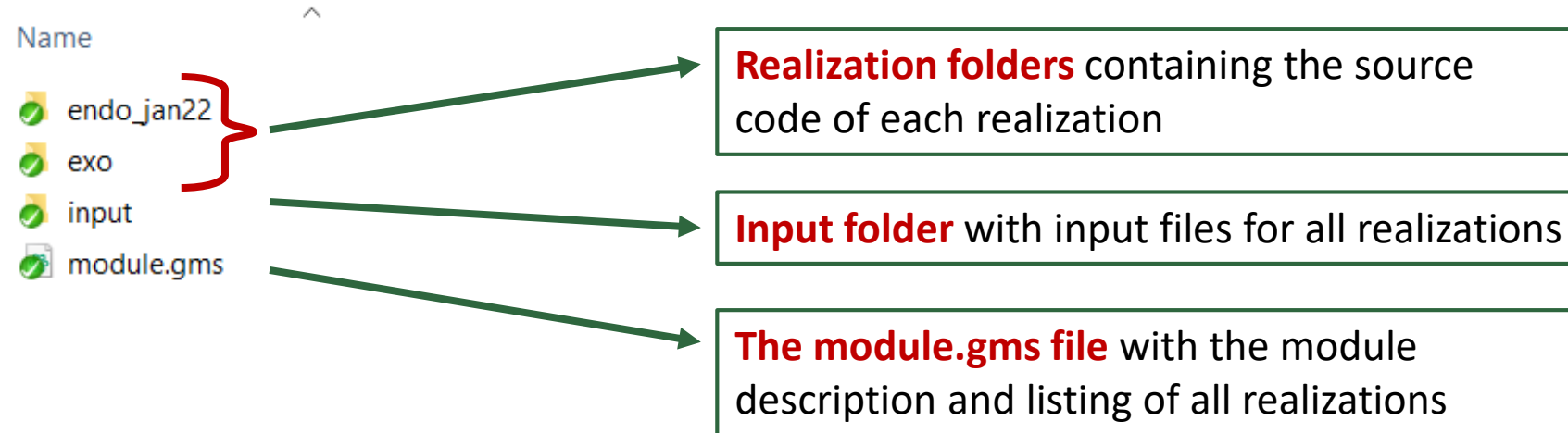
The configuration settings are set in the **default.cfg** file (or in the start scripts).

Each folder within the **modules** folder represents a component of the model and contains different realizations (approaches to that component).

Structure of the modules

Each module (within the modules folder) has a similar structure:

MAGPIE Workshop 2026 > WS26 model version > magpie > modules > 13_tc



New realizations can be added by keeping the same structure. In that sense, MAGPIE is easily extendable.

Structure of realizations



As with the modules folders, the **realizations files (.gms) and folders are similarly constructed**. The source code is distributed over several gms-files to ensure the correct order of calculations during the optimization:

MAGPIE Workshop 2026 > WS26 model version > magpie > modules > 13_tc > endo_jan22

 declarations.gms	11 Jun 2026 16:04	GMS File
 equations.gms	11 Jun 2026 16:04	GMS File
 input.gms	11 Jun 2026 16:04	GMS File
 nl_fix.gms	11 Jun 2026 16:04	GMS File
 nl_relax.gms	11 Jun 2026 16:04	GMS File
 nl_release.gms	11 Jun 2026 16:04	GMS File
 postsolve.gms	11 Jun 2026 16:04	GMS File
 preloop.gms	11 Jun 2026 16:04	GMS File
 presolve.gms	11 Jun 2026 16:04	GMS File
 realization.gms	11 Jun 2026 16:04	GMS File
 scaling.gms	11 Jun 2026 16:04	GMS File
 sets.gms	11 Jun 2026 16:04	GMS File

Note that not every gms-file is needed in every realization.

gms.file	function
declarations.gms	Declares all variables, equations, and parameters for the realization.
equations.gms	Contains functional relationships that have to be fulfilled within the optimization.
input.gms	Loads input from <code>any_module/input</code> or <code>any_module/a_realization/input</code> .
sets.gms	Lists sets that are used (mainly) within this realization or are needed for interfaces defined within this realization.
preloop.gms	Includes calculations to be executed before the time-step loop starts.
presolve.gms	Includes calculations to be executed for each time step before the optimization process.
postsolve.gms	Includes calculations to be executed for each time step after the optimization process and defines output.
nl_fix.gms	Fixes non-linear behaviour to linear behavior.
nl_release.gms	Releases restrictions imposed during non-linear fixes.
scaling.gms	Lists the expected order of magnitude of specific variables calculated in this realization to improve the efficiency of the run.
not_used.txt	Lists interfaces (declared in other modules) that are not used within this realization, but in other realizations of the same module.

Coding etiquette: Variable and parameter naming



Prefixes are extended using either **m** (used in multiple modules/core code) or a **two digit number** (only used in the current module).

```
1  *** | (C) 2008-2025 Potsdam Institute for Climate Impact Research (PIK)
2  *** | authors, and contributors see CITATION.cff file. This file is part
3  *** | of MAGPIE and licensed under AGPL-3.0-or-later. Under Section 7 of
4  *** | AGPL-3.0, you are granted additional permissions described in the
5  *** | MAGPIE License Exception, version 1.0 (see LICENSE file).
6  *** | Contact: magpie@pik-potsdam.de
7
8  equations
9  q38_cost_prod_labor(i)      Regional labor input costs for crop production (mio USD17MER)
10 q38_cost_prod_capital(i)    Regional capital input costs for crop production (mio USD17MER)
11 q38_investment_immobile(j,kr) Cellular immobile investments into farm capital (mio USD17MER)
12 q38_investment_mobile(j)    Cellular mobile investments into farm capital (mio USD17MER)
13 ;
14
15 positive variables
16 vm_cost_prod_crop(i,factors) Regional factor costs of capital and labor for crop production (mio USD17MER per yr)
17 v38_investment_immobile(j,kr) Investment costs in immobile farm capital (mio USD17MER per yr)
18 v38_investment_mobile(j)     Investment costs in mobile farm capital (mio USD17MER per yr)
19 ;
20
21 parameters
22 p38_labor_need(t,i,kr)       Labor input costs per unit of output (USD17MER per ton DM)
23 p38_capital_need(t,i,kr,mobil38) Capital requirements per unit of output (USD17MER per ton DM)
24 p38_capital_immobile(t,j,kr) Preexisting immobile capital stocks before investment (mio USD17MER)
25 p38_capital_mobile(t,j)     Preexisting mobile capital stocks before investment (mio USD17MER)
26
27 p38_capital_cost_shares_iso(t,iso) Capital shares of factor costs on iso level (1)
28 p38_capital_share_calibration(iso) Summation factor used to calibrate calculated capital shares with historical values (1)
29 pm_factor_cost_shares(t,i,factors) Capital and labor shares of factor costs on regional level (1)
30
31 p38_croparea_start(j,w,kr)    Agricultural land initialization area (mio. ha)
32
33 i38_fac_req(t_all,i,kr)       Factor requirements (USD17MER per tDM)
34 ;
35
36
37 ##### R SECTION START (OUTPUT DECLARATIONS) #####
38 parameters
39 ov_cost_prod_crop(t,i,factors,type) Regional factor costs of capital and labor for crop production (mio USD17MER per yr)
40 ov38_investment_immobile(t,j,kr,type) Investment costs in immobile farm capital (mio USD17MER per yr)
41 ov38_investment_mobile(t,j,type)     Investment costs in mobile farm capital (mio USD17MER per yr)
```

q_ Equations

v_ Variables

p_ processing parameters

i_ input parameters

o_ output parameters

... In other files you will also find:

c_ Switches

s_ Scalars

f_ File parameters

m_ Macros

Coding etiquette: Variable and parameter naming



In other cases, the **prefixes are extended** with **a second letter** to indicate details such as:

?c_ current time step

```
pcm_land(j,land)
```

?q_ parameter containing the values of an equation

```
oq10_land(t,j,type)
```

?v_ parameter containing the values of a variable

```
ov_landreduction(t,j,land,type)
```

Suffixes indicate the **level of aggregation** of an object:

- (no suffix) Highest disaggregation available
- **_setname** aggregation **over set**
- **_reg** **regional** aggregation
- **_glo** **global** aggregation

```
i42_wat_req_k(t,j,k)
```

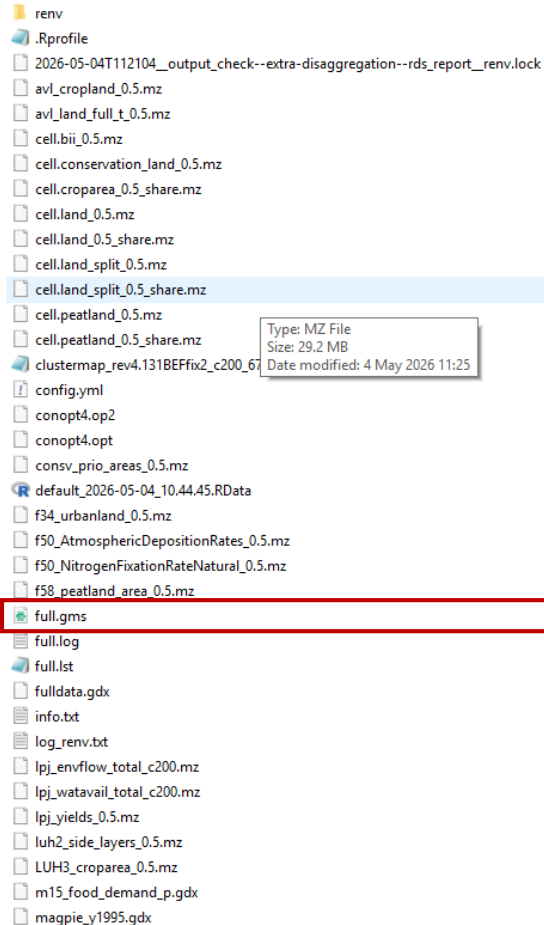
```
v11_cost_reg(i)
```

```
i32_max_aff_area_glo
```

The full.gms file



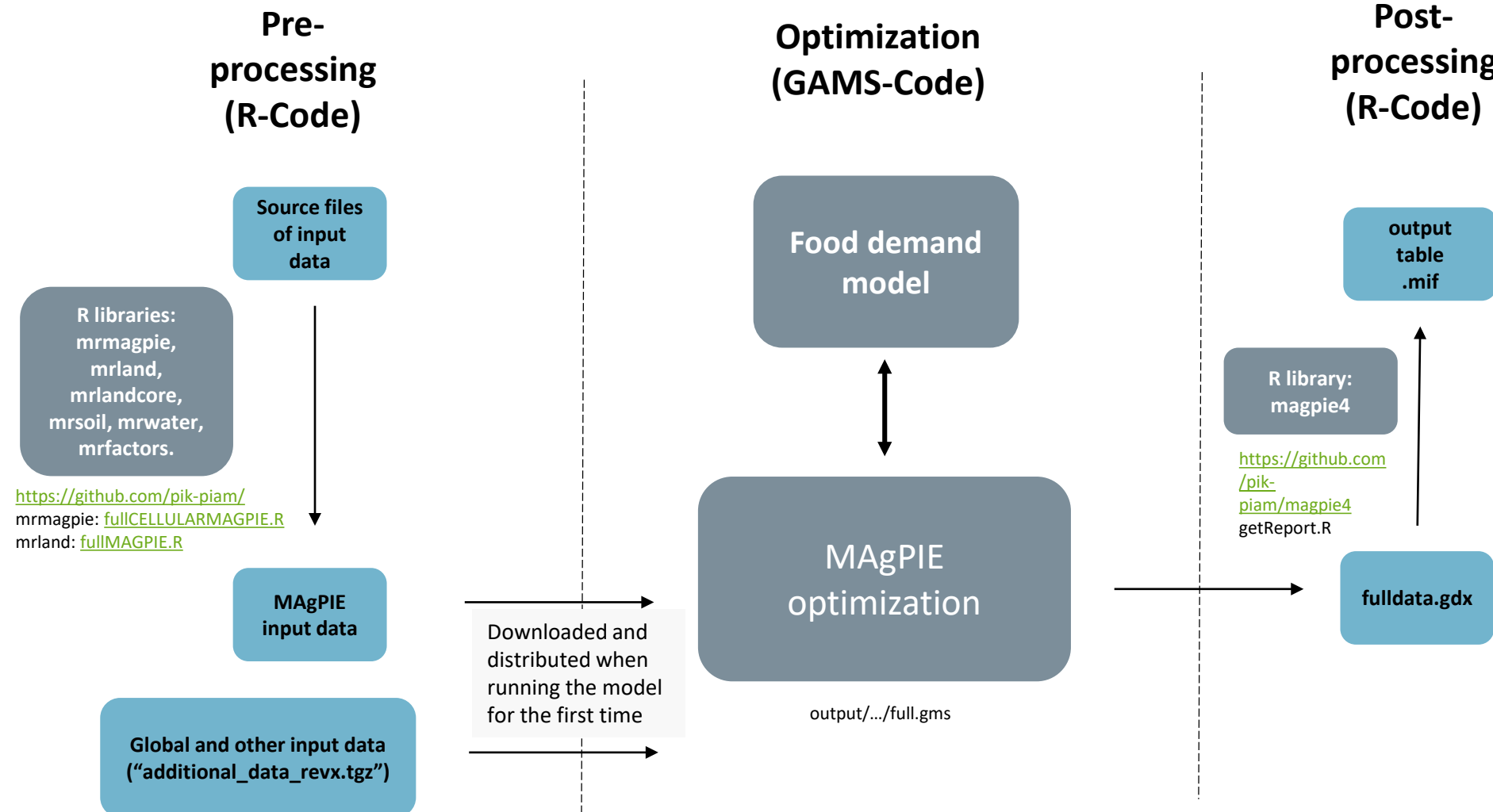
Once you have started a magpie run, you can find the full.gms file in the **ouput folder**.



full.gms

Contains the final code used in the current MAgPIE run, based on the selected settings and on one realization per module (created automatically).

MAGPIE environment: pre- and postprocessing



Brief exercise



(For online participants: Green tick when you are ready) 

1. Within the magpie folder, find the **13_tc** module.
2. Open the **declaration.gms** file of the **endo_jan22** realization.
3. Find one example of:
 - A parameter used only within the current module
 - An equation
 - A variable used in this module and the core code/another module
 - An output parameter

Thank you